SUMMER INTERNSHIP

GenAI Powered Data Analytics

Tata iQ – Job Simulation via Forage

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1. Internship Overview

Tata – GenAI Powered Data Analytics Job Simulation

Financial-services case focused on AI-powered data analytics and strategy development.

Used GenAI-assisted analysis to assess data quality, identify risk indicators and structure predictive-modelling logic.

Designed an AI-driven collections strategy using agentic AI, automation and responsible-AI principles.

Connected analytical findings to practical business recommendations.

2. Exploratory Data Analysis

Understanding the Geldium customer dataset

500 customer records across 19 columns; 16% of customers were marked delinquent (80/500).

Missing values: Income – 39; Credit_Score – 2; Loan_Balance – 29.

No duplicate rows and no duplicate Customer_ID values were identified.

Candidate risk indicators: recent payment behaviour, missed payments, credit utilization, DTI, credit score, income and loan balance.

Employment_Status labels require normalization before modelling.

3. Key Data Insights

Findings used to shape the modelling approach

Simple linear correlations with delinquency were weak, so candidate predictors require formal validation.

Missing Loan_Balance group showed ~24.1% observed delinquency versus ~15.5% where the field was present.

Credit_Utilization values above 1.0 should be validated for unit/scale consistency.

The 16% positive class means evaluation should not rely on accuracy alone.

4. Predictive Modelling Framework

Initial no-code / interpretable modelling plan

Primary model: Logistic Regression for an interpretable probability of delinquency.

Prepare data through imputation, category normalization, encoding and appropriate scaling.

Use stratified train/test splitting and cross-validation.

Evaluate precision, recall, F1, AUC-ROC, confusion matrix and calibration.

Use a Decision Tree as a challenger; introduce complexity only when validated improvement justifies it.

5. Business Recommendation

Turning risk prediction into early intervention

Run a 12-week early-outreach pilot for customers assessed as higher risk.

Use supportive reminders, repayment guidance and appropriate repayment arrangements.

SMART target: 10% relative reduction in 30+ day delinquency versus a comparable control group.

Review pilot performance at weeks 4, 8 and 12.

Thresholds should reflect customer impact, intervention cost and collections capacity.

6. AI-Powered Collections Strategy

End-to-end intervention workflow

DATA INPUT → RISK SCORE + BUSINESS RULES → ACTION → OUTCOME MONITORING

Automated actions can include reminders, general repayment guidance and low-risk follow-ups.

The system should capture outcomes and feed them into a learning/monitoring loop.

Human staff handle sensitive, complex or high-impact situations.

7. Agentic AI + Human Oversight

Automation with controlled decision boundaries

Agentic AI: automate routine reminders, general offers and low-risk follow-up activities.

Human oversight: tailored debt restructuring, legal escalation, complaints, high-risk cases and sensitive scenarios.

Maintain clear escalation rules, action logs and review points.

AI should support decision-making rather than replace human judgement in critical cases.

8. Responsible AI Guardrails

Fairness, explainability, privacy and compliance

- Monitor group-level precision, recall and false-positive/false-negative rates.
- Review potential disparate impact and proxy variables such as location or employment-related fields.
- Provide understandable reasons for risk flags and intervention recommendations.
- Protect customer data through appropriate privacy, access-control and governance practices.
- Apply human-in-the-loop review for critical decisions and comply with applicable requirements.

9. Skills Developed & Conclusion

Learning outcomes

AI analytics • AI strategy • EDA • data quality management • predictive analytics

Model selection • model validation • analytical reporting • business communication

Ethical reasoning • regulatory compliance • process automation • strategic thinking

Key takeaway: effective AI requires reliable data, validated models, responsible automation and measurable business impact.